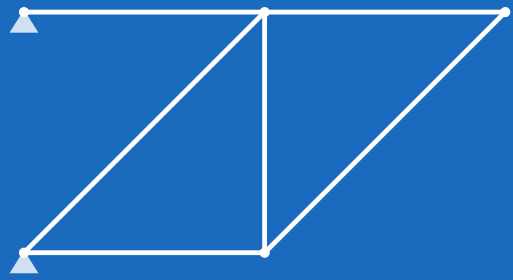
TRUSS REPORT

Wall bracket (cantilever)

Plane truss — analysis and CIRSOC 301 rod checks

Author A. Student · Course Structures I · 12/10/2026



MAX TENSION	MAX COMPRESSION	DEFLECTION	MOST USED ROD	BUCKLING Λ_1	WEIGHT
+50.0 kN	**-35.4 kN**	**4.5 mm**	**0.76**	**1.69**	**178 kg**
rod 0	rod 3	L/1348 · limit L/300	rod 5 · elastic buckling (E3-3	the load can grow ×1.7	carries 25.0 kN · 14× own weight

The model

Span	6.00 m
Height	3.00 m
Nodes / rods	5 / 6
Supports	pin at 0, pin at 1
Loads	1 nodal, total 25.0 kN
Family "Chord"	CHS 101.6x4.0, F24 (CIRSOC) / A36
Family "Web"	CHS 76.1x3.6, F24 (CIRSOC) / A36

What the analysis found

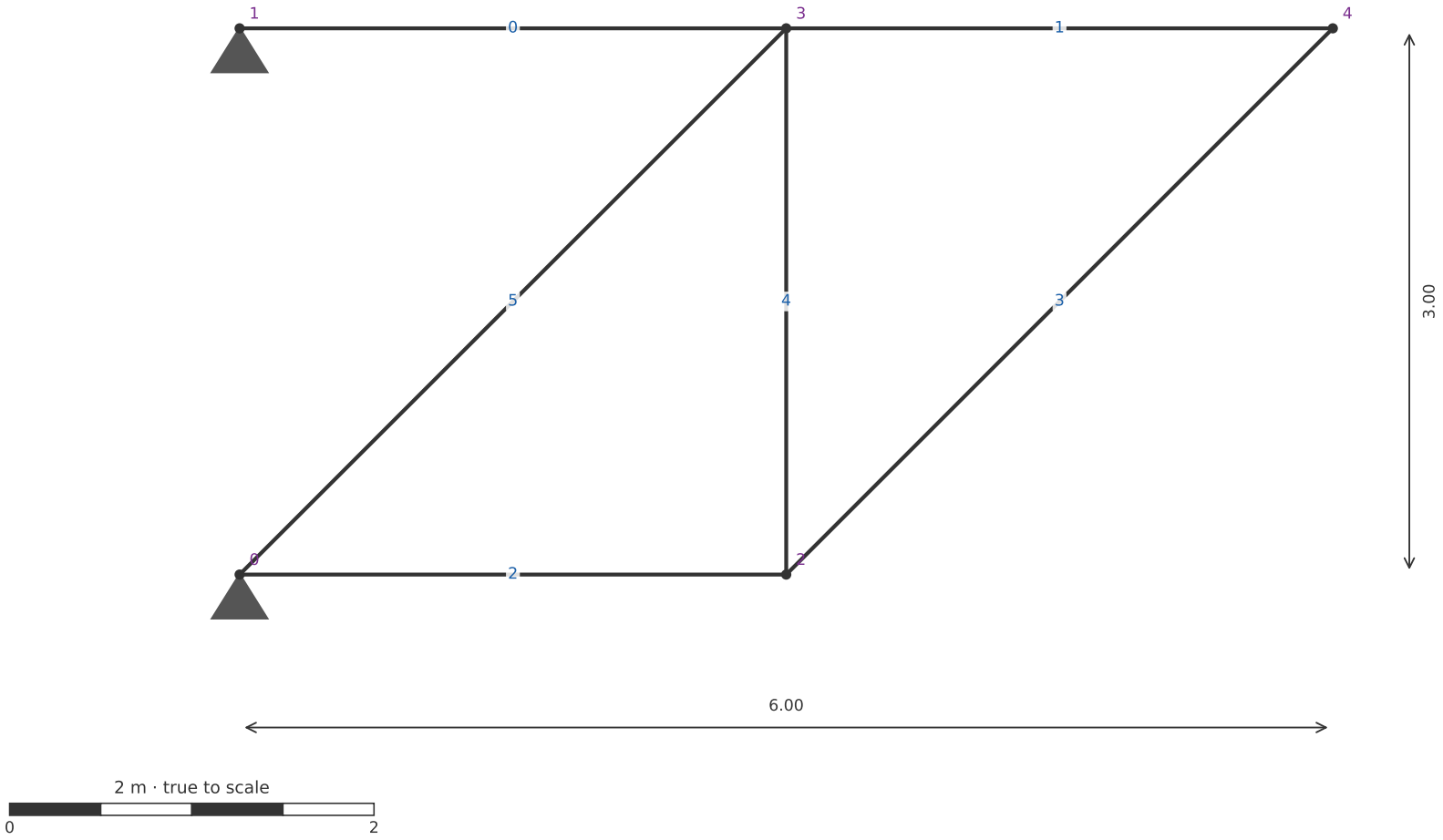
Stability	Statically determinate (m + r − 2j = 6 + 4 − 2×5 = 0): statics alone gives every force.
Most used rod	rod 5 at 0.76 (elastic buckling (E3-3))
Rods over capacity	0 of 6 checked
Buckling	$\lambda_1 = 1.69$
Shear and moment	none: every rod is pinned and loaded only at its nodes, so the axial force is the whole result

Contents

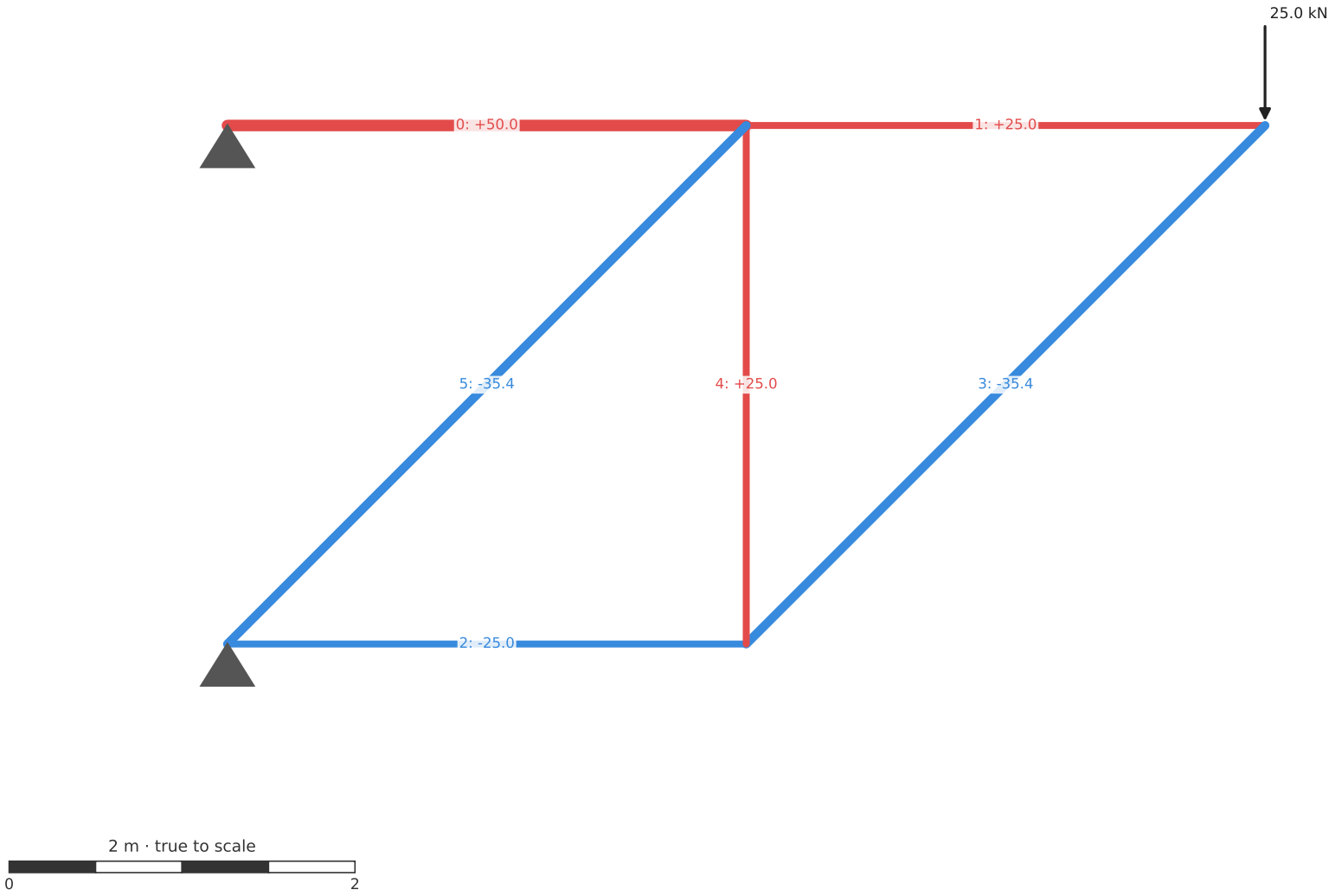
2 Geometry
3 Axial forces
4 Deflection and stress
5 Reactions
6 Utilisation
7 Buckling
8 Rods
9 Explain rod 5 and 3

Results are an aid to a qualified professional, who must check them. Rod checks to CIRSOC 301 as implemented; first-order analysis; joints and connections are not checked here.

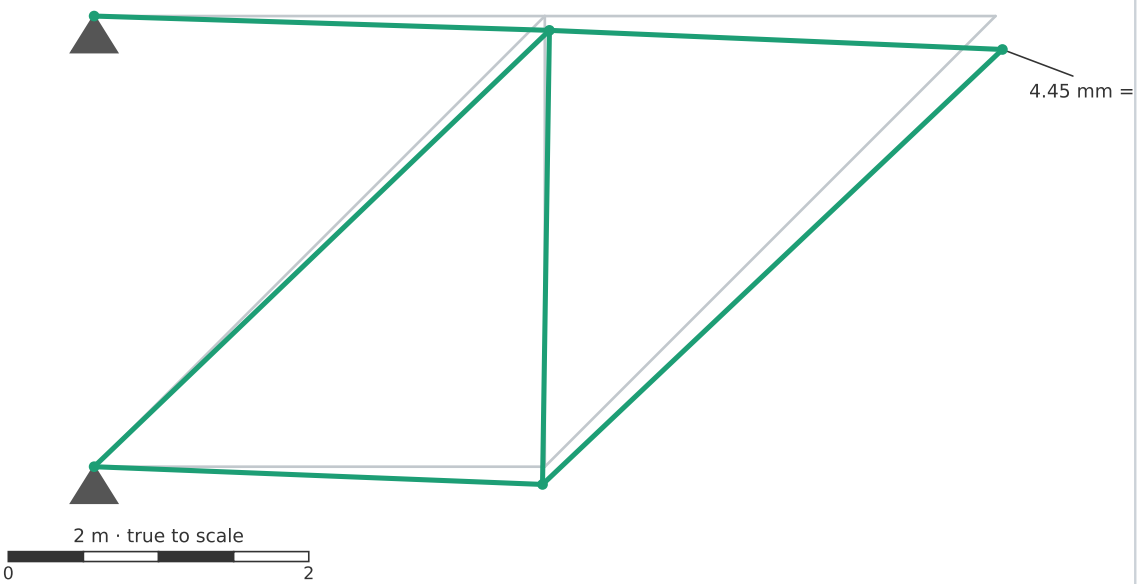
Node numbers in purple, rod numbers in blue. Dimensions in m.



Axial force N (kN): red pulls (+), blue pushes (−), grey carries nothing; line width follows |N| (largest 50.0).

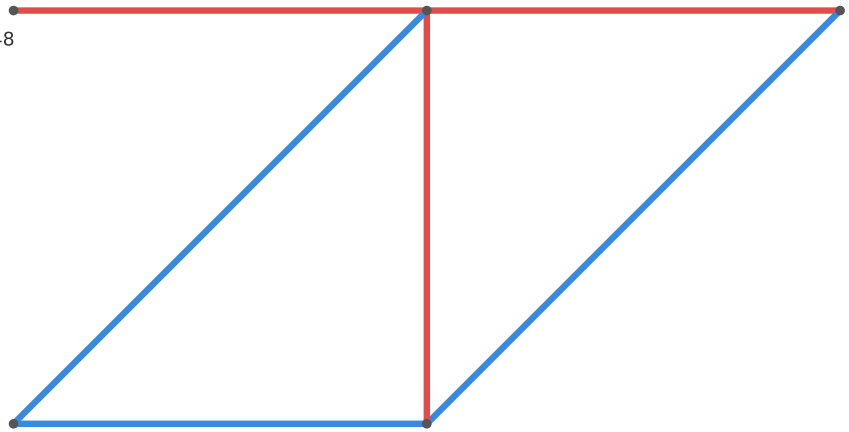


Elastic line: the deflected shape (green) over the truss (grey), every rigid rod drawn as the curve it bends into. Geometry to scale; displacements magnified × 50. Every rod is pinned, so every member stays straight: there is no curvature, and no contraflexure. Largest vertical deflection 4.45 mm against the limit L/300 = 20.0 mm: OK.

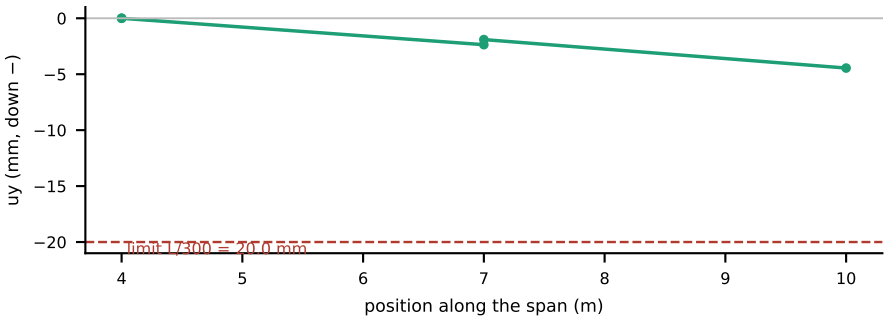


Fibre stress top and bottom face of each member

red = tension, blue = compression. Every rod is pinned, so both faces of a rod carry the same stress and each rod is one colour: the axial force.

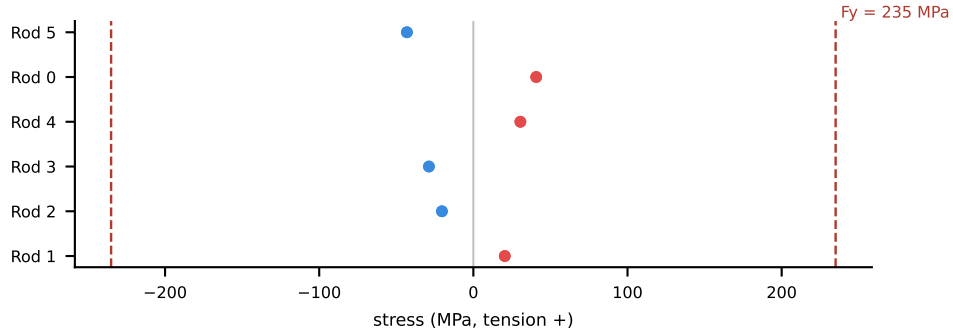


Deflection along the span vertical movement of every joint



Largest 4.45 mm at joint 4 = L/1348. Within the limit.

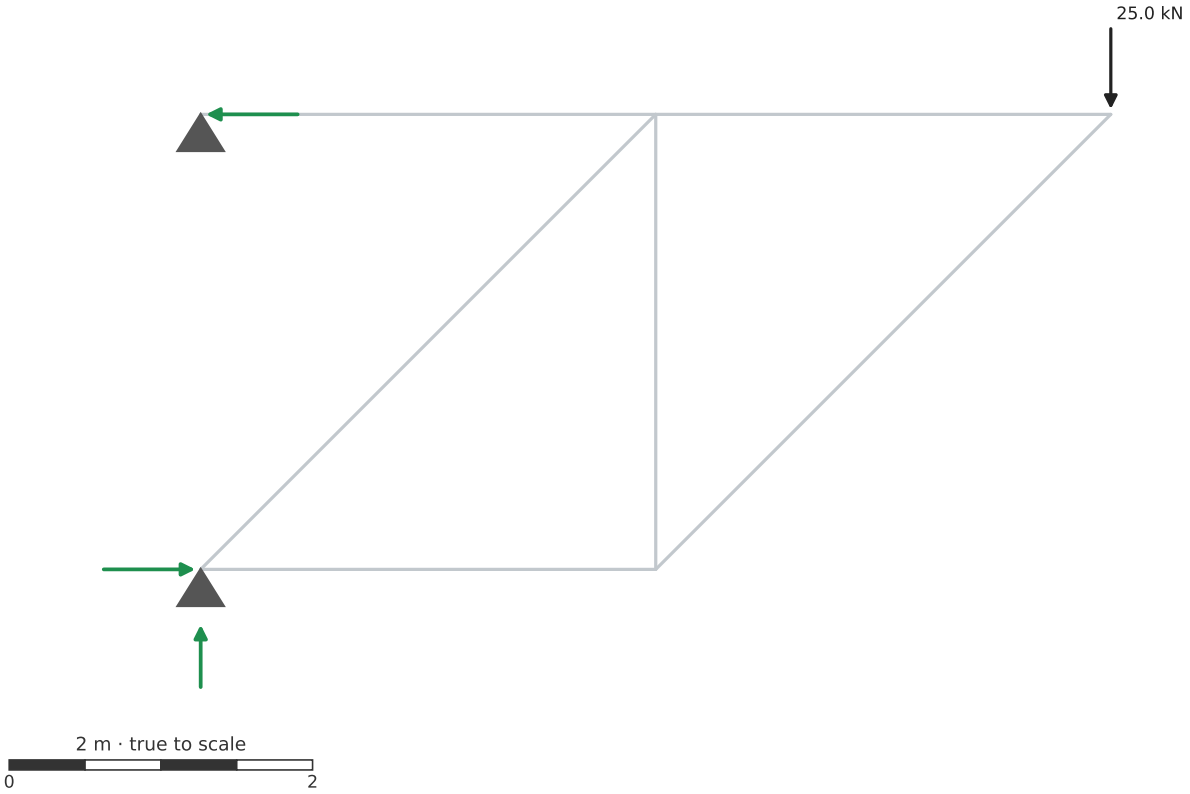
Fibre stress values the two faces of each rod, MPa



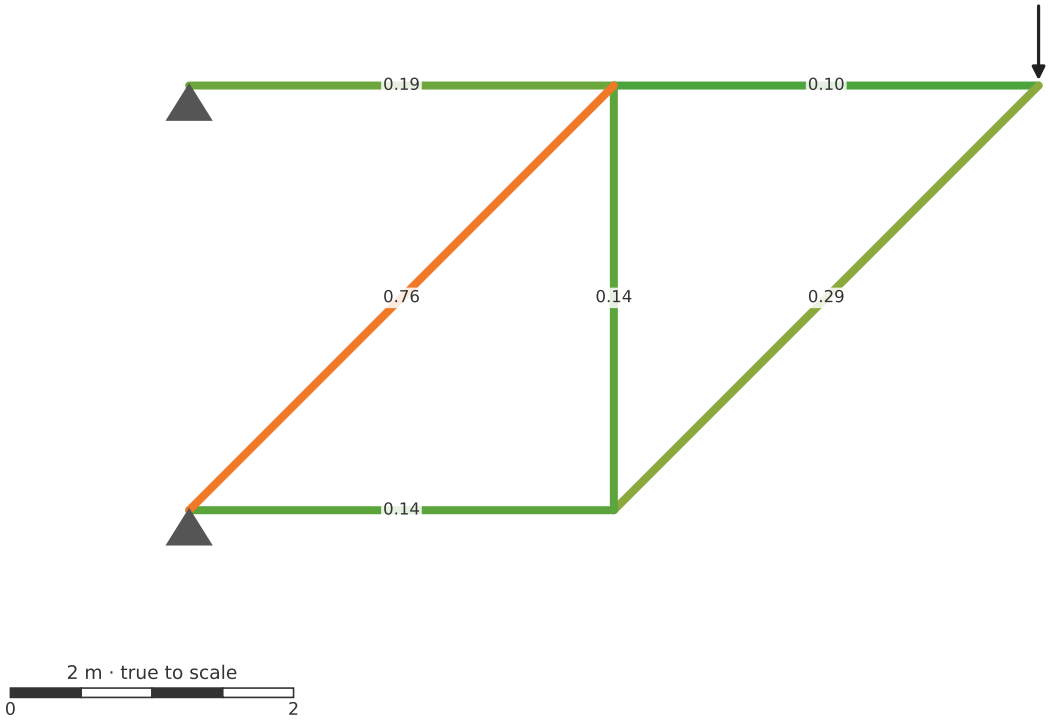
Every rod is pinned, so its two fibres carry the same N/A and coincide: no bending opens the pair.

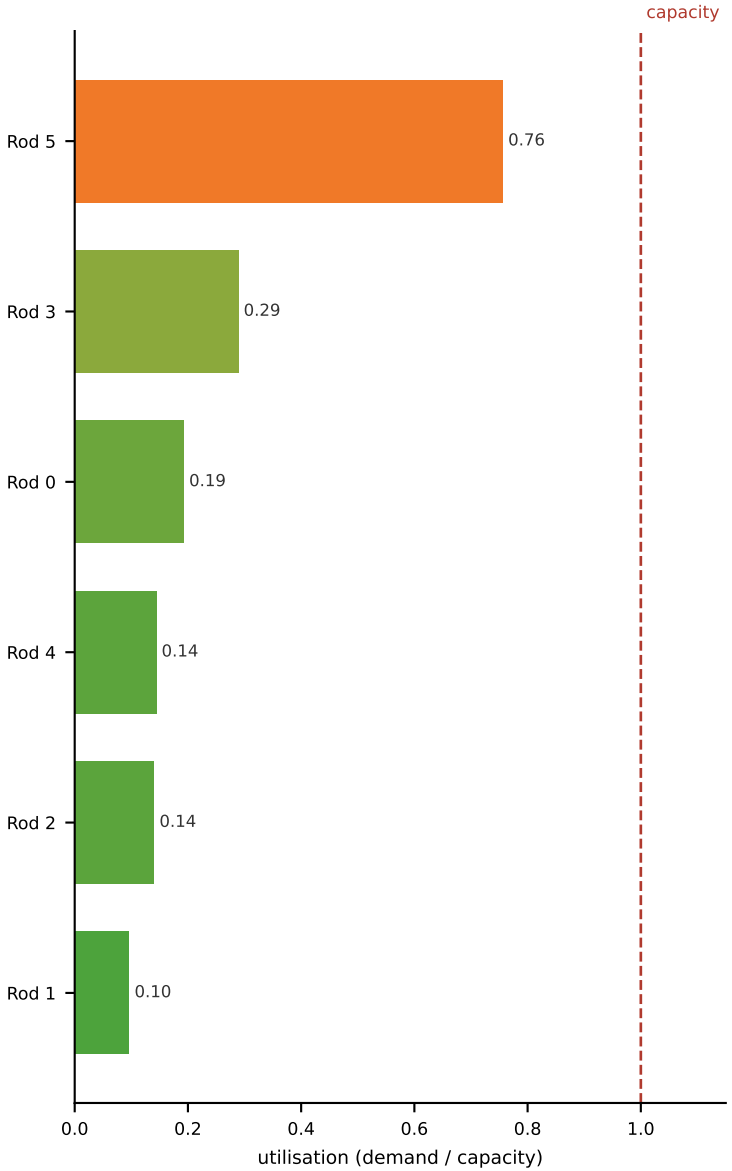
Support reactions (green arrows), as the supports push on the truss; Ry positive upwards. The sums balance the loads.

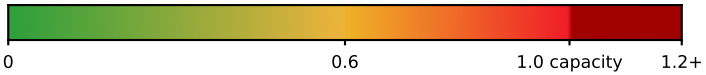
Node	Type	Rx (kN)	Ry (kN)
0	pin	50.00	25.00
1	pin	-50.00	0.00
Σ		0.00	25.00



Utilisation: demand over CIRSOC 301 capacity. Most used rod 5 at 0.76 (elastic buckling (E3-3)). Every checked rod is within capacity.



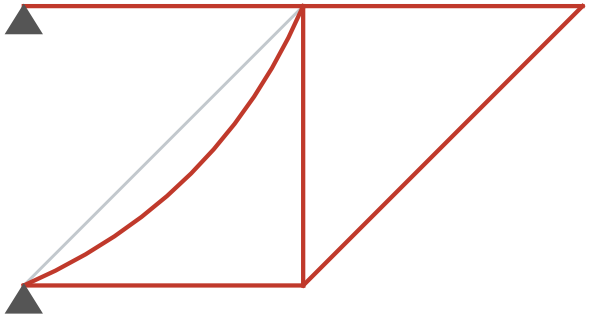




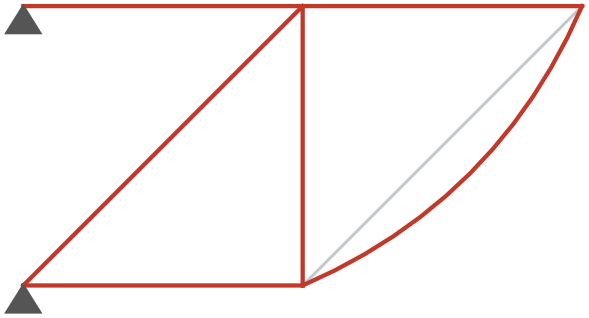
grey = no steel section, not checked

λ is the factor on the load as it stands. $\lambda_1 = 1.69$, so the load can grow 1.7× before it buckles. Mode 2 is 4.57, 171 % higher: one member decides the buckling load.

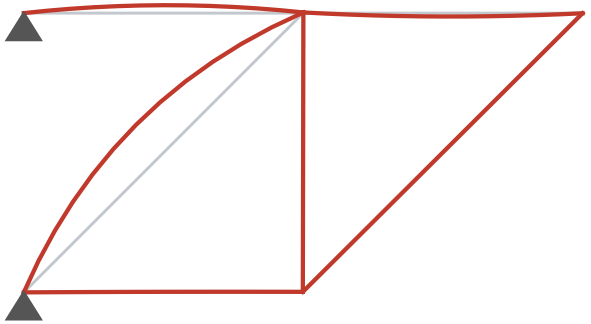
Mode 1: λ = 1.69
rod 5 bows

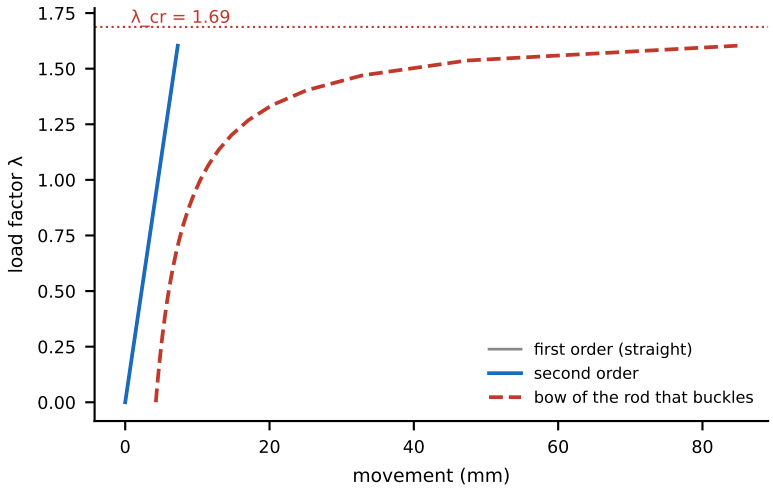


Mode 2: λ = 4.57
rod 3 bows



Mode 3: λ = 8.15
rod 5 bows



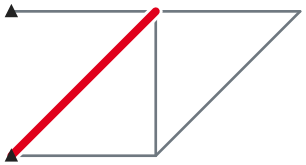


The second-order line bends away from the first-order one and runs off as λ nears λ_cr: that gap IS the amplification every second-order design rule is about. Started from an initial bow of L/1000 = 4.24 mm in the first mode's shape.

Elastic buckling of the IDEAL truss, so every λ is an UPPER bound: imperfections, residual stresses and yielding are not modelled, and each rod is cut in two so it can bow between its joints. Buckling OUT of the plane is the rod check's job (CIRSOC 301 E3, minor radius) and is in the Utilisation and Rods sheets, not here. A mode has a shape, not a size: the shapes are drawn exaggerated.

Every rod: its end nodes, family and steel section, length, axial force N (+ pulls, − pushes), the CIRSOC 301 check that governs it and how much of its capacity that uses (Use = demand / capacity; OVER above 1.0).

Rod	Nodes	Family	Section	L (m)	N (kN)	Governing check	Use
0	1–3	Chord	CHS 101.6x4.0	3.00	+50.00	normal stress (H.3.4)	0.19 OK
1	3–4	Chord	CHS 101.6x4.0	3.00	+25.00	normal stress (H.3.4)	0.10 OK
2	0–2	Chord	CHS 101.6x4.0	3.00	-25.00	inelastic buckling (E3-2)	0.14 OK
3	2–4	Chord	CHS 101.6x4.0	4.24	-35.36	inelastic buckling (E3-2)	0.29 OK
4	2–3	Web	CHS 76.1x3.6	3.00	+25.00	normal stress (H.3.4)	0.14 OK
5	0–3	Web	CHS 76.1x3.6	4.24	-35.36	elastic buckling (E3-3)	0.76 OK



Rod 5

utilisation 0.76

elastic buckling (E3-3)

```
Rod 5
=====
Steel, CIRSOC 301-2018 -- A = 820 mm², Fy = 235 MPa, L = 4.243 m

 1. Slenderness  [E2]
      K·L / r
      = 1 × 4,243 mm / 25.7 mm
      = 165.3

 2. Euler stress  [E3-4]
      Fe = π² E / (KL/r)²
      = π² × 200,000 / 165.3²
      = 72.23 MPa

 3. Which branch  [E3]
      KL/r ≤ 4.71 √(E/Fy) ?
      = 165.3 vs 4.71 √(200,000/235) = 137.4
      = elastic

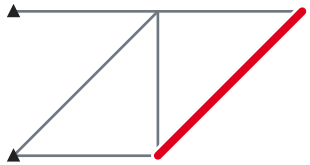
 4. Critical stress  [E3-3]
      Fcr = 0.877 · Fe
      = 0.877 × 72.23
      = 63.35 MPa

 5. Design strength  [E3-1]
      Pd = φc · Fcr · A
      = 0.9 × 63.35 × 820
      = 46.75 kN

 6. Compression ratio  [E3]
      |N| / Pd
      = 35.36 / 46.75
      = 0.756

 7. Governs
      the largest ratio
      = elastic buckling (E3-3)
      = 0.756

utilisation 0.756 -- OK
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Rod 3

utilisation 0.29

inelastic buckling (E3-2)

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Rod 3
=====
Steel, CIRSOC 301-2018 -- A = 1,226 mm², Fy = 235 MPa, L = 4.243 m

 1. Slenderness  [E2]
      K·L / r
      = 1 × 4,243 mm / 34.5 mm
      = 122.8

 2. Euler stress  [E3-4]
      Fe = π² E / (KL/r)²
      = π² × 200,000 / 122.8²
      = 130.80 MPa

 3. Which branch  [E3]
      KL/r ≤ 4.71 √(E/Fy) ?
      = 122.8 vs 4.71 √(200,000/235) = 137.4
      = inelastic

 4. Critical stress  [E3-2]
      Fcr = 0.658^(Fy/Fe) · Fy
      = 0.658^(235/130.80) × 235
      = 110.78 MPa

 5. Design strength  [E3-1]
      Pd = φc · Fcr · A
      = 0.9 × 110.78 × 1,226
      = 122.29 kN

 6. Compression ratio  [E3]
      |N| / Pd
      = 35.36 / 122.29
      = 0.289

 7. Governs
      the largest ratio
      = inelastic buckling (E3-2)
      = 0.289

utilisation 0.289 -- OK
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